

Outline for Proof by Contradiction

Proposition P .

Proof. Suppose $\sim P$.

$\vdots$

Therefore $C \wedge \sim C$. ■

Note:

Proof of " $\forall x \in D, P(x) \Rightarrow Q(x)$ " by contradiction:

Let $\forall x \in D, P(x) \Rightarrow Q(x)$ -----(i)

Assume that (i) is **false**.

Then, $\sim(\forall x \in D, P(x) \Rightarrow Q(x))$.

i.e. $\exists x_0 \in D, P(x_0) \wedge \sim Q(x_0)$.

$\left. \begin{array}{l} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \right\} \begin{array}{l} \text{Definitions,} \\ \text{Axioms,} \\ \text{Logical equivalences,} \\ \text{Rules of inference} \end{array}$

..... is a **contradiction**.

Hence, (i) is **not false**. (i.e. Our assumption is **false**).

Thus, $\forall x \in D, P(x) \Rightarrow Q(x)$. ■

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Proof of " $\forall x \in D, P(x)$ " by contradiction:

Let $\forall x \in D, P(x)$ -----(i)

Assume that (i) is **false**.

Then, $\sim(\forall x \in D, P(x))$.

i.e. $\exists x_0 \in D, \sim P(x_0)$.

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..... is a **contradiction**.

Hence, (i) is **not false**. (i.e. Our assumption is **false**).

Thus, $\forall x \in D, P(x)$. ■

Example 1. for all integers n , if n^2 is odd, then n is odd

Example 2. For every $n \in \mathbb{Z}$, $4 \nmid (n^2 + 2)$.

Prf! Let for all integers n , if n^2 is odd, then n is odd — (i)

Suppose (i) is false

Then there exists at least $n_0 \in \mathbb{Z}$ such that n_0^2 is odd and n_0 is not odd
i.e. n_0 is even $\therefore n_0 = 2k$ for some $k \in \mathbb{Z}$

Note:

$\therefore n_0^2 = 2m, m = 2k^2 \in \mathbb{Z}$
 $\therefore n_0^2$ is even
 $\therefore n_0^2$ is not odd

This is **contradiction**
Hence, (i) is **true** ■

Some important properties of integers

1] Addition Properties.

- The set of integers is *closed under addition*. For any $a, b \in \mathbb{Z}$, $a + b \in \mathbb{Z}$.
- Addition is *commutative*. For any $a, b \in \mathbb{Z}$, $a + b = b + a$.
- Addition is *associative*. For any $a, b, c \in \mathbb{Z}$, $(a + b) + c = a + (b + c)$.
- There is a *zero element* $0 \in \mathbb{Z}$, satisfying $0 + a = a = a + 0$ for any $a \in \mathbb{Z}$.
- For any $a \in \mathbb{Z}$, there exists an *additive inverse* $-a \in \mathbb{Z}$ satisfying $a + (-a) = 0 = (-a) + a$.

Definition: Subtraction in $\mathbb{Z}$ is defined by $a - b = a + (-b)$ for $a, b \in \mathbb{Z}$.

2] Multiplication Properties.

- a) $\mathbb{Z}$ is *closed under multiplication*. For any $a, b \in \mathbb{Z}$, $a \cdot b \in \mathbb{Z}$.
- b) Multiplication is *commutative*. For any $a, b \in \mathbb{Z}$, $ab = ba$.
- c) Multiplication is *associative*. For any $a, b, c \in \mathbb{Z}$, $(ab)c = a(bc)$.
- d) There is an *identity element* $1 \in \mathbb{Z}$ satisfying $1 \cdot a = a = a \cdot 1$ for any $a \in \mathbb{Z}$.

3] **Distributive property.** This is the one property that combines both addition and multiplication. For any $a, b, c \in \mathbb{Z}$, $a(b + c) = ab + ac$. One can deduce the additional distributive laws, $(a+b)c = ac+bc$, $a(b-c) = ab-ac$ and $(a-b)c = ac-bc$.

4] **Trichotomy Principle.** The set of integers can be partitioned into three disjoint sets, $\mathbb{Z} = -\mathbb{N} \cup \{0\} \cup \mathbb{N}$, where

$\mathbb{N} = \{1, 2, 3, \dots\} = \text{Natural Numbers} = \text{Positive Integers}$,

$-\mathbb{N} = \{-1, -2, -3, \dots\} = \text{Negative Integers}$.

One then defines the inequalities $>$ and $<$ by saying $a > b$ if $a - b \in \mathbb{N}$ and $a < b$ if $a - b \in -\mathbb{N}$. Thus we get the Law of Trichotomy which states that for any two integers a, b exactly one of the following holds: $a < b$, $a = b$ or $a > b$, (that is $a - b \in -\mathbb{N}$, $a - b = 0$ or $a - b \in \mathbb{N}$.)

5] **Positivity.** The sum of two positive integers is positive. The product of two positive integers is positive.

Definition. Let a and b be integers. We say that a divides b , and write $a \mid b$, if there is an integer q such that $b = a \cdot q$. In this case, we also say that b is divisible by a and call a and q divisors or factors of b .

If a does not divide b , we write $a \nmid b$.

Definition. The greatest common divisor (GCD) of two nonzero integers a and b is the largest of all common divisors of a and b . It is denoted by $\gcd(a, b)$. If $\gcd(a, b) = 1$, then the integers a and b are said to be coprime (or relatively prime).

Definition. Let $n > 1$ be an integer. Then n is called a prime if 1 and n are the only positive divisors.

We say that n is composite if it has divisor d with $1 < d < n$.

Note:

The numbers greater than 1 that are not prime are called *composite* numbers.

It follows immediately from the definition that, if a prime p does not divide a nonzero integer a , then $\gcd(p, a) = 1$, that is, p and a are coprime.

Exercise:

1. Prove that $\frac{1}{2}$ is not an integer.
2. Prove that $\frac{1}{3}$ is not an integer.
3. Let $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. Prove that if p and q are coprime, then $p^2 \neq 2q^2$.

If-and-Only-If Proof (Biconditional proof)

Some propositions have the form

P if and only if Q .

We know that this statement asserts that **both** of the following conditional statements are true:

If P , then Q .

If Q , then P .

So to prove " P if and only if Q ," we must prove **two** conditional statements.

Recall that $Q \Rightarrow P$ is called the *converse* of $P \Rightarrow Q$.

Thus we need to prove both $P \Rightarrow Q$ and its converse.

These are both conditional statements, so we may prove them with either direct, contrapositive or contradiction proof. Here is an outline:

Outline for If-and-Only-If Proof

Proposition P if and only if Q .

Proof.

[Prove $P \Rightarrow Q$ using direct, contrapositive or contradiction proof.]

[Prove $Q \Rightarrow P$ using direct, contrapositive or contradiction proof.] ■

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$$p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$$
$$p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (\sim p \Rightarrow \sim q)$$

Example 1: The integer n is odd if and only if n^2 is odd.

Example 2. for all integers n , n is even if and only if n^3 is even.

Example 3. For every integer n , $6 \mid n$ iff $2 \mid n$ and $3 \mid n$.

Proof. Let n be an arbitrary integer.

(Ex)

Example 4. Let a be an integer. Prove that the following are equivalent.

- a is divisible by 3.
- $3a$ is divisible by 9.
- $a + 3$ is divisible by 3.

Mid –Exams

8 June 2026, 22 June 2026, 13 July 2026

First year 3.00-4.00pm

End of D11